Restore IP Addresses (medium)

Problem Statement

Given a string `s` containing `digits` only, return a list of string containing the valid `ip addresses` that can be formed by inserting the `dots(.)` in between string characters.

A `valid IP address` consists of four parts, each ranging from `0` to `255`, separated by dots. It's important to note that leading zeros in any part of the IP address are not allowed, except for the number 0 itself.

- **For example**, `"1.1.21.1"`, and `"1.12.1.1"` are `valid` ip addresses, and `"1.1.021.1"` is invalid ip address.

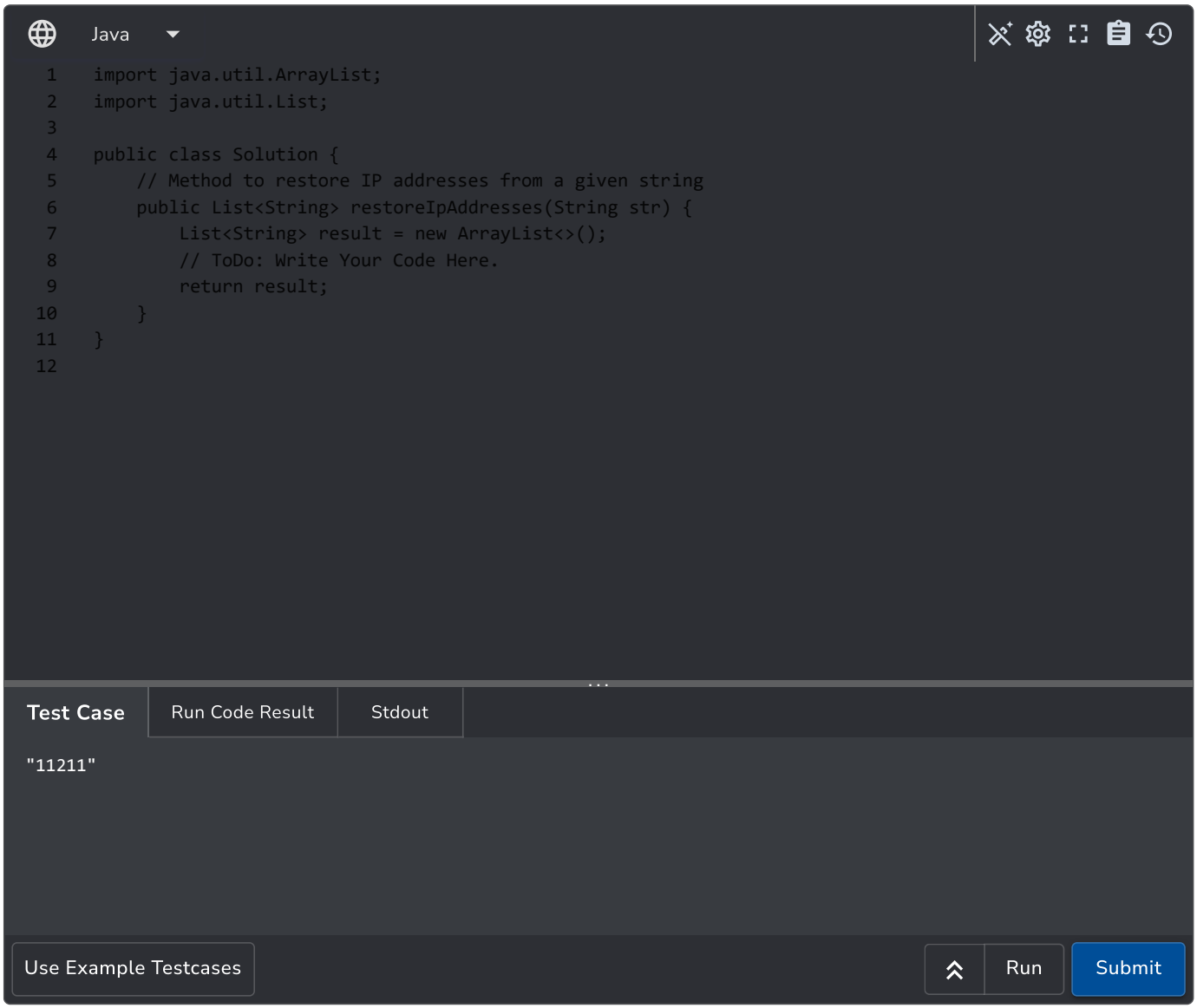
Examples

- **Example 1**:
  - **Input**: `"11211"`
  - **Expected Output**: `["1.1.2.11", "1.1.21.1", "1.12.1.1", "11.2.1.1"]`
  - **Justification**: These are all the valid ways to insert dots into the input string such that each segment forms a valid part of an IP address without leading zeros, except for the number 0 itself, and each segment is within the range of 0 to 255.
- **Example 2**:
  - **Input**: `"1111"`
  - **Expected Output**: `["1.1.1.1"]`
  - **Justification**: Based on the string's length and value, there is only one way to form a valid IP address by inserting dots after each digit.
- **Example 3**:
  - **Input**: `"0000"`
  - **Expected Output**: `["0.0.0.0"]`

- **Justification**: Despite leading zeros being generally disallowed, in this case, each segment of the IP address is the number 0 itself, making it the only valid formation.

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